

Liver Transplant, Pediatric

ORG: P-795 (ISC)

[Link to Codes](#)**MCG Health**Inpatient & Surgical
Care
28th Edition

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Care Planning - Inpatient Admission and Alternatives

Clinical Indications for Procedure

- Procedure is indicated for **ALL** of the following(1)(2)(3)(4)(5)(6)(7):
 - ☐ Transplant needed for **1 or more** of the following:
 - ☐ Acute liver failure (also known as "fulminant liver failure"), as indicated by **ALL** of the following(1)(8)(9):
 - Acute liver disease (ie, 8 weeks' duration or less)[A]
 - Hepatic encephalopathy of any degree[B]
 - Severe manifestations of liver disease, as indicated by **1 or more** of the following:
 - Acute need for mechanical ventilation
 - Acute need for renal replacement therapy
 - International normalized ratio (INR) of 2.0 or more
 - ☐ Severe chronic liver disease, as indicated by **1 or more** of the following(2):
 - Model for End-Stage Liver Disease (MELD) or Pediatric Model for End-Stage Liver Disease (PELD) score of 15 or higher[C][D][E][F]
 - ☐ Complication of liver disease, as indicated by **1 or more** of the following:
 - Severe growth failure (eg, z score is less than 2 standard deviations below mean for age and sex)
 - Parenteral nutrition necessary to maintain growth or euglycemia
 - Ascites or hepatic hydrothorax that persists or recurs despite diuresis and low-sodium diet
 - Hepatic encephalopathy that persists despite treatment (eg, lactulose, rifaximin)
 - Recurrent spontaneous bacterial peritonitis
 - Recurrent severe cholangitis (eg, 2 or more episodes within 6 months requiring intravenous antibiotics for 2 weeks via central access)
 - History of life-threatening infection (eg, bacteremia, hypotension) with need for ICU stay
 - Variceal bleeding that persists despite treatment (eg, beta-blocker therapy, band ligation)
 - Biliary atresia(12)
 - Hepatocellular carcinoma with no evidence of tumor outside the liver(2)(13)(14)
 - ☐ Cholangiocarcinoma and **ALL** of the following(1)(15)(16):
 - Tumor less than or equal to 3 cm in diameter

- Unresectable due to location (eg, perihilar, vital vessel involvement), extent (eg, multifocal, bilobar disease), or severe underlying hepatic disease
- No metastatic or nodal disease
- Neoadjuvant chemoradiation planned prior to transplant
- Hepatoblastoma without metastases that is not amenable to hepatic resection(1)(2)(17)
- Epithelioid hemangioendothelioma unresponsive to therapy that is not amenable to hepatic resection(2)
- [-] Neuroendocrine tumor and **ALL** of the following(2)(17):
 - Primary tumor and extrahepatic disease resected with curative intent 6 months or more prior to transplant
 - Stable disease for 6 months or more prior to transplant
 - Tumor not amenable to resection
 - No evidence of extrahepatic metastases
- [-] Polycystic liver disease and **ALL** of the following(18):
 - Surgical therapy (removal of cysts, fenestration) not effective or not possible
 - Severe disease, as indicated by **1 or more** of the following:
 - Hepatic decompensation (eg, ascites, hepatic encephalopathy)
 - Concurrent hemodialysis or GFR less than 20 mL/min/1.73m² (0.33 mL/sec/1.73m²)
 -  eGFR - Pediatric Calculator
 - Prior kidney transplant
 - Moderate to severe protein calorie malnutrition
 - Severe sarcopenia
- [-] Primary sclerosing cholangitis and **ALL** of the following(19)(20)(21):
 - Cirrhosis on imaging or liver biopsy
 - Transplant necessary due to **1 or more** of the following:
 - Biliary tract stricture not responsive to intervention
 - Intractable pruritus
 - 2 or more episodes of cholangitis
- Familial amyloidosis(1)
- Urea cycle disorder(1)(22)
- Organic acidemia(1)(23)
- [-] Hepatopulmonary syndrome[G] and **ALL** of the following(1):
 - Sequelae of portal hypertension (eg, ascites, varices, splenomegaly, thrombocytopenia)
 - Arterial partial pressure of oxygen of less than 60 mm Hg (8.0 kPa) (on room air) not otherwise explained by underlying pulmonary disease
 - Right-to-left cardiac shunt demonstrated by contrast echocardiogram ("bubble study") or nuclear perfusion lung scan
- [-] Portopulmonary hypertension and **ALL** of the following(1)(26):
 - Portal hypertension
 - Pulmonary hypertension on right heart catheterization with **ALL** of the following(27)(28):
 - Mean pulmonary artery pressure greater than 35 mm Hg at rest
 - Pulmonary vascular resistance 3 Wood units (240 dynes*sec/cm⁵) or greater
 - Other causes of pulmonary hypertension not a significant contributor to pulmonary hypertension (eg, heart failure, valvular heart disease, parenchymal pulmonary disease)
 - Satisfactory response to pulmonary hypertension treatment (eg, vasodilators, endothelial receptor blocker), as evidenced by **1 or more** of the following[H](25):
 - Post-treatment mean pulmonary artery pressure is less than 35 mm Hg and pulmonary vascular resistance is less than 5 Wood units (400 dynes*sec/cm⁵).
 - Post-treatment mean pulmonary artery pressure is more than 35 mm Hg and less than 45 mm Hg and pulmonary vascular resistance is less than 3 Wood units (240 dynes*sec/cm⁵).
- Acute decompensated Wilson disease(1)(29)
- Anhepatic state[I]
- Acute hepatic porphyria with recurrent intractable symptoms despite pharmacotherapy (eg, prophylactic hemin or givosiran therapy)(31)
- Failure of previous liver transplant(1)(7)
- Hepatic artery thrombosis in transplanted liver(1)(7)
- Other hepatic disorder associated with increased mortality that can be addressed with liver transplant (eg, glycogen storage disease, familial intrahepatic cholestasis, severe trauma, alpha-1 antitrypsin deficiency)[E](2)(7)(32)
- No contraindications to liver transplant present

Alternatives to Procedure

- Alternatives include(5)(6):
 - Continuing medical management of liver disease(8)(29)(33)(34)(35)(36)(37)
 - For acetaminophen or other drug toxicity: N-acetylcysteine and discontinuation of hepatotoxic agent(33)(38)(39)

- For bleeding esophageal varices: portacaval shunt, transjugular intrahepatic portosystemic shunt, endoscopic variceal ligation or sclerotherapy(40)(41)(42)(43)
- For biliary atresia: hepatoportoenterostomy (Kasai procedure)(5)(44)
- Resection for single early hepatocellular carcinoma(14)(45)
- Resection for hepatoblastoma
- Resection for selected non-hepatocellular carcinomas or liver metastases(15)(46)
- Ablation, radiation, or arterially directed therapies (eg, chemoembolization, radioembolization) for hepatocellular carcinoma, cholangiocarcinoma, or liver metastases(15)(45)(47)
- Systemic chemotherapy or immunotherapy (eg, immune checkpoint inhibitor, angiogenesis inhibitor, tyrosine kinase inhibitor) for advanced hepatocellular carcinoma or cholangiocarcinoma(15)(45)
- For primary sclerosing cholangitis: anion-exchange resins, antipruritic agents, endoscopic dilation for dominant strictures(20)(21)
- For acute hepatic porphyria: avoidance of triggers, analgesic agents and antiemetics as needed, and prophylactic hemin or givosiran therapy(31)
- Palliative care(15)(45)(48)

Operative Status Criteria

Goal Length of Stay: 10 days postoperative

- Inpatient

Preoperative Care Planning

- Preoperative care planning needs may include(5)(6)(7)(32):
 - Preoperative evaluation, including:
 - Evaluation by multidisciplinary liver transplant team for:
 - Severity of liver failure
 - Alternatives to liver transplant
 - Contraindications to liver transplant
 - Psychosocial evaluation
 - Protocol as specified by transplant center
 - Diagnostic test scheduling, including:
 - Laboratory testing
 - Immunology screening, including blood and tissue typing
 - Detailed infectious disease screening, including viral antibody titers(49)
 - Nutrition evaluation(50)
 - Dental evaluation
 - Cardiac evaluation, including echocardiogram (eg, for portopulmonary hypertension or hepatopulmonary syndrome)(25)
 - Right heart catheterization (for portopulmonary hypertension)
 - Pulmonary function testing (eg, for cystic fibrosis patient)
 - CT or MRI
 - Serum alpha-fetal protein for hepatocellular carcinoma
 - Bone scan for hepatocellular carcinoma
 - Routine preoperative evaluation. See Preoperative Education, Assessment, and Planning Tool [SR](#).
 - Preoperative treatment, procedures, and stabilization, including:
 - Nutritional optimization(50)
 - Aggressive ascites treatment to decrease associated high-energy expenditures
 - Vaccination(49)
 - Liver biopsy(45)
 - For hepatorenal syndrome: albumin transfusion, IV fluids, somatostatin analogue, and possible renal replacement therapy[J](51)
 - For cholangiocarcinoma: neoadjuvant chemoradiation(2)(16)(52)
 - Addressing all treatable coexisting conditions before transplant
 - Preoperative discharge planning as appropriate. See Discharge Planning in this guideline.

Hospitalization

Optimal Recovery Course

Day	Level of Care	Clinical Status	Activity	Routes	Interventions	Medications
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1	• OR to ICU • Social Determinants of Health Assessment • Discharge planning	• Clinical Indications met[K] • Possible postoperative extubation[L]	• Bed rest	• IV fluids • IV medications	• Central line • CBC • Liver function and coagulation studies	• Perioperative antibiotics • Immunosuppressive medications and corticosteroids • Prophylactic antiemetic for postoperative nausea and vomiting prophylaxis[M] • Multimodal analgesia, possible PCA or epidural analgesics
2	• ICU • Social Determinants of Health Assessment	• Extubated • Hypotension absent	• Bed rest	• IV fluids • IV medications • Possible liquid diet	• CBC • Liver function and coagulation studies	• Discontinue antibiotics • Immunosuppressive medications and corticosteroids • Multimodal analgesia
3	• ICU • Social Determinants of Health Assessment	• Hemodynamic stability • No evidence of biliary dysfunction • Acidosis absent	• Bed rest	• IV fluids • IV medications • Advance diet	• CBC • Liver function and coagulation studies	• Immunosuppressive medications and corticosteroids • Multimodal analgesia
4	• ICU • Possible floor • Social Determinants of Health Assessment	• No evidence of ileus or bowel obstruction • Mental status at baseline or improved • Hypoxemia absent	• Bed rest	• Liquid diet • IV fluids • IV medications	• Laboratory tests	• Immunosuppressive medications and corticosteroids • Multimodal analgesia
5	• Floor • Social Determinants of Health Assessment	• Pain absent or managed • Pleural effusion absent or improved	• Bed rest	• IV fluids • IV medications • Advance diet	• Laboratory tests	• PCA or epidural analgesics absent[N] • Immunosuppressive medications and corticosteroids
6	• Floor • Social Determinants of Health Assessment	• Electrolyte abnormalities absent or improved • Renal function at baseline or improved	• Up to chair[O]	• Usual diet • IV fluids • IV medications	• NG tube absent • Laboratory tests	• Immunosuppressive medications and corticosteroids

7	- Floor - Social Determinants of Health Assessment	No evidence of rejection	Limited ambulation[O]	- IV fluids - IV medications - Usual diet	Laboratory tests	Immunosuppressive medications and corticosteroids
8	- Floor - Social Determinants of Health Assessment	Anemia absent or acceptable for next level of care	Ambulatory	Oral hydration Oral medications - Usual diet	Laboratory tests	Immunosuppressive medications and corticosteroids
9	- Floor - Social Determinants of Health Assessment	No evidence of postoperative or surgical site infection	Ambulatory	- Oral hydration - Oral medications - Usual diet	Central lines absent - Laboratory tests	Immunosuppressive medications and corticosteroids
10	- Floor - Social Determinants of Health Assessment	Laboratory tests confirm adequate liver function and coagulation Pleural effusion absent or improved	Ambulatory	- Oral hydration - Oral medications - Usual diet	Laboratory tests	Immunosuppressive medications and corticosteroids
11	- Social Determinants of Health Assessment - Floor to discharge[P] - Complete discharge planning	Hemodynamic stability Laboratory tests confirm adequate liver function and coagulation Hypoxemia absent No evidence of postoperative or surgical site infection No evidence of biliary dysfunction Mental status at baseline Electrolyte abnormalities absent or acceptable for next level of care Pleural effusion absent or acceptable for next level of care Renal function at baseline or acceptable for next level of care	Ambulatory or acceptable for next level of care[Q]	Oral hydration[R] Oral medications or regimen acceptable for next level of care Oral diet or acceptable for next level of care	Laboratory tests	Immunosuppressive medications and corticosteroids

- | | | | | | |
|--|---|--|--|--|--|
| | <ul style="list-style-type: none"> • No evidence of rejection • Pain absent or managed • Discharge plans and education understood | | | | |
|--|---|--|--|--|--|

(4)(6)(50)(55)(56)(57)N

Recovery Milestones are indicated in **bold**.

Goal Length of Stay: 10 days postoperative

Note: Goal Length of Stay assumes optimal recovery, decision making, and care. Patients may be discharged to a lower level of care (either later than or sooner than the goal) when it is appropriate for their clinical status and care needs.

Extended Stay

Minimal (a few hours to 1 day), Brief (1 to 3 days), Moderate (4 to 7 days), and Prolonged (more than 7 days).

- Extended stay beyond goal length of stay may be needed for(6)(57)(58):
 - Failure to meet discharge criteria (recovery milestones within final day in Optimal Recovery Course)
 - Expect brief stay extension.
 - Continued Hypotension
 - Expect brief stay extension.
 - Hemorrhage, portal vein thrombosis, pancreatitis(6)(59)
 - Postoperative hemorrhage may require factor replacement, transfusion, and possible re-exploration.
 - Portal vein thrombosis may require anticoagulation, endovascular intervention, or surgical exploration.
 - Expect brief to moderate stay extension.
 - Biliary complications, leaks(59)(60)
 - Small bile leaks may heal spontaneously.
 - Anticipate imaging studies and ERCP with stenting or percutaneous transhepatic biliary drainage, or surgical intervention for large or nonresolving bile leak.
 - Expect moderate stay extension.
 - Continued metabolic abnormalities (eg, hyponatremia, hypernatremia, electrolyte disorder, hypervolemia)
 - Expect brief to moderate stay extension.
 - Damage control surgery if severe intraoperative difficulty or complications upon reperfusion (eg, requiring deferred completion of biliary reconstruction)
 - Expect moderate stay extension.
 - Acute hepatic necrosis
 - Expect brief to moderate stay extension.
 - Very severe preoperative liver disease
 - Higher preoperative Model for End-Stage Liver Disease (MELD/PELD) scores (eg, above 20) correlate with longer lengths of stay.
 - Anticipate brief to moderate stay extension.
 - Liver transplant performed in neonatal period (eg, secondary to congenital malformation)
 - Expect brief to moderate stay extension.
 - Acute rejection or graft dysfunction causing mild to moderate liver failure(61)
 - Expect brief to moderate stay extension.
 - Hepatic artery thrombosis(59)(60)(61)
 - Anticipate imaging studies and surgical or interventional radiology-guided revascularization, or possible retransplant.
 - Expect brief to moderate stay extension.
 - Cardiac complications or dysfunction(62)
 - Pulmonary edema, arrhythmia may occur.
 - Anticipate diuresis, antiarrhythmic agent, vasopressor agent, or possible cardiac intervention.
 - Expect brief to moderate stay extension.
 - Pulmonary complications(62)(63)
 - Pleural effusion, atelectasis, acute respiratory distress syndrome, postoperative hepatopulmonary syndrome with Hypoxemia, portopulmonary hypertension, transfusion-related acute lung injury, or pneumonia may complicate recovery.
 - Anticipate thoracentesis, chest physiotherapy, possible pulmonary vasodilator or antibiotic therapy, and possible need for noninvasive or invasive mechanical ventilation.

- Expect brief to moderate stay extension.
- Severe nutritional deficiency(50)
 - Expect brief stay extension.
- Acute kidney injury or hepatorenal syndrome(64)
 - Patients with worse preoperative synthetic function (eg, elevated INR) are at higher risk of postoperative acute kidney injury.
 - Expect brief stay extension.
- Infectious complications(65)
 - Expect brief to moderate stay extension.
- Continued postoperative ileus
 - Expect brief stay extension.
- Altered mental status(66)
 - Expect brief to moderate stay extension.

See Common Complications and Conditions [ISC](#) for further information.

Hospital Care Planning

- Hospital evaluation and care needs may include(4)(5)(6)(32):
 - Diagnostic test scheduling and completion, including:
 - Laboratory tests, including CBC, prothrombin time, partial thromboplastin time, and liver function tests
 - Abdominal ultrasound
 - Treatment and procedure scheduling and completion, including(50):
 - IV antibiotics
 - Establishment of immunosuppressive and corticosteroid regimen: monitor levels and observe for toxicity
 - Early oral or enteral nutrition
 - Parenteral nutrition if oral or enteral feeding is contraindicated (eg, severe hepatic encephalopathy) or not practical. See Malnutrition: Common Complications and Conditions [ISC](#) for further information.
 - Nutritional support
 - Multimodal analgesia
 - Consultation, assessment, and other services scheduling and completion, including multidisciplinary transplant team
 - Monitoring patient's status for deterioration and comorbid conditions (see Inpatient Monitoring and Assessment Tool [SR](#)); key items include:
 - Hemodynamic stability
 - Closely monitor fluid balance to avoid hypervolemia.
 - Respiratory status
 - Clinical manifestations of rejection, including:
 - Fever and tachycardia
 - Change in color, amount, and consistency of bile drainage through T-tube
 - Elevation of liver enzymes and increasing jaundice
 - Right upper quadrant or flank pain
 - Clinical manifestations of venous thromboembolism. See Venous Thrombosis and Pulmonary Embolism: Common Complications and Conditions [ISC](#) for further information.
 - Renal function. See Renal Failure: Common Complications and Conditions [ISC](#) for further information.
 - Pain management, including PCA
 - Wound management, observing for healing and signs of infection
 - T-tube management
 - Observe for bile leakage or signs of peritonitis.

Discharge

Discharge Planning

- Discharge planning includes[S]:
 - Assessment of needs and planning for care, including(68)(69):
 - Develop and modify treatment plan (involving multiple providers) as needed.
 - Evaluate and address preadmission functioning as needed.
 - Evaluate and address psychosocial status issues as indicated. See Psychosocial Assessment [SR](#) for further information.
 - Evaluate and address social determinants of health (eg, housing, food). See Social Determinants of Health Screening Tool [SR](#) for further information.(67)
 - Evaluate and address patient or caregiver preferences as indicated.

- Identify skilled services needed at next level of care, with specific attention to:
 - Medication management, adherence instruction, and side effects assessment(4)(70)(71)
 - Pain management(72)
 - T-tube management(73)
 - Wound or dressing management(73)
- Early identification of anticipated discharge destination; options include(69)(74):
 - Home, considerations include:
 - Home safety assessment. See Home Safety Assessment [SR](#) for further information.
 - ☐ Patient safe to go home; examples include(75)(76)(77):
 - Medical status stable for patient's condition
 - Functional care can safely be provided with available resources.
 - Mental status stable for patient's condition
 - Medication availability confirmed and reconciliation complete
 - Patient/caregiver education completed with written discharge instructions provided
 - Community resources identified and referrals made, as needed
 - Home care arranged, if indicated
 - Necessary medical equipment delivery arranged or available in home, if indicated
 - Necessary medical supplies ordered, or patient/caregiver can obtain, if indicated
 - Access to follow-up care
 - Self-management ability if appropriate. See Activities of Daily Living (ADL) and Instrumental Activities of Daily Living (IADL) Assessment [SR](#) for further information.
 - Caregiver need, ability, and availability
 - Post-acute skilled care or custodial care as indicated. See Discharge Planning Tool [SR](#) for further information.
- Transitions of care plan complete, including(69)(74)(78):
 - Patient and caregiver education complete. See Liver Transplant, Pediatric: Patient Education for Clinicians [SR](#) for further information.
 - See Teach Back Tool [SR](#) for further information.
 - ☐ Medication reconciliation completion includes(71)(79):
 - Compare patient's discharge list of medications (prescribed and over-the-counter) against provider's admission or transfer orders.
 - Assess each medication for correlation to disease state or medical condition.
 - Report medication discrepancies to prescribing provider, attending physician, and primary care provider, and ensure accurate medication order is identified.
 - Provide reconciled medication list to all treating providers.
 - Confirm that patient or caregiver can acquire medication.
 - Educate patient and caregiver.
 - Provide complete medication list to patient and caregiver.
 - Importance of presenting personal medication list to all providers at each care transition, including all provider appointments
 - Reason, dosage, and timing of medication (eg, use "teach-back" techniques)(80)
 - Encourage communication between patient, caregiver, and pharmacy for obtaining prescriptions, setting up home medication delivery, and reviewing for drug-drug interactions.
 - See Medication Reconciliation Tool [SR](#) for further information.
 - Plan communicated to patient, caregiver, and all members of care team, including(81):
 - Inpatient care and service providers
 - Primary care provider
 - All post-discharge care and service providers
 - Appointments planned or scheduled, which may include:
 - Primary care provider
 - General surgeon
 - Specialists for management of comorbidities as needed(73)
 - Transplant center or team
 - Other
 - Outpatient testing and procedure plans made, which may include:
 - Laboratory testing(4)
 - Other
 - Referrals made for assistance or support, which may include:
 - Community services
 - Educational program (eg, chronic condition management, self-management)(82)
 - Financial, for follow-up care, medication, and transportation

- Self-help or support groups(73)
- Tobacco use treatment(73)
- Other
- Medical equipment and supplies coordinated (ie, delivered or delivery confirmed), which may include:
 - Blood pressure cuff
 - Child safety seat(83)
 - Drain care supplies
 - Nutritional supplements
 - Weight scale
 - Wound care equipment and supplies(84)
 - Other

Discharge Destination

- Post-hospital levels of admission may include:
 - Home.
 - Home healthcare. See Home Care Indications for Admission Section [HC](#) in Transplant, Pediatric guideline in Home Care.

Evidence Summary

Criteria

The evidence for the clinical indications found in this guideline includes 19 published peer reviewed articles, 4 specialty society or other evidence-based guidelines, and 5 book sections.

Length of Stay

Analysis of national hospital discharge data for a pediatric population shows 25% of hospitalized patients undergoing liver transplant discharged in 10 days or less.(57) (**EG 3**)

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Footnotes

[A] The standard definition of acute liver failure in children is acute liver disease of less than 8 weeks' duration plus either an INR greater than 2, or an INR greater than 1.5 with any degree of hepatic encephalopathy.(10) However, for purposes of liver transplant, the Organ Procurement and Transplantation Network defines acute liver failure (also known as "fulminant liver failure") as the onset of hepatic encephalopathy within 8 weeks of the first signs or symptoms of liver disease.(1) [A in Context Link 1]

[B] Hepatic encephalopathy is graded by West Haven criteria for patients age 3 years and older, or by the Whittington Scale for patients younger than 3 years of age.(10)(11) A calculator for West Haven criteria is available at <https://www.mdcalc.com/hepatic-encephalopathy-grades-stages>. Whittington Scale criteria for early encephalopathy (stage I and II) include inconsolable crying, sleep reversal, and inattention to task; moderate encephalopathy (stage III) is characterized by stupor, combativeness, and somnolence with hyperreflexia; and late encephalopathy (stage IV) is characterized by coma (either arousable to painful stimuli (IVa) or with no response to pain (IVb)) with absent reflexes and decerebrate or decorticate posturing.(10) [B in Context Link 1]

[C] The Model for End-Stage Liver Disease (MELD) score is used for liver allocation for patients 12 years of age and older; an individual's score can be calculated at the Organ Procurement and Transplantation Network website: <http://optn.transplant.hrsa.gov/converge/resources/MeldPeldCalculator.asp?index=98>.(1) [C in Context Link 1]

[D] The Pediatric Model for End-Stage Liver Disease (PELD) score, a model similar to the Model for End-Stage Liver Disease (MELD), has been developed for children 11 years of age and younger. An individual's score can be calculated at the Organ Procurement and Transplantation Network website: <http://optn.transplant.hrsa.gov/converge/resources/MeldPeldCalculator.asp?index=99>.(1) [D in Context Link 1]

[E] In cases where the calculated Model for End-Stage Liver Disease (MELD) or Pediatric Model for End-Stage Liver Disease (PELD) score does not reflect the urgent need for a liver transplant, an appeal may be made to the Regional Review Board for a score exception.(1)(2) Examples of conditions eligible for MELD or PELD score exclusions include hepatopulmonary syndrome, primary

hyperoxaluria, metabolic disorders, hepatocellular carcinoma, and cystic fibrosis with FEV-1 less than 40% predicted.(1)(2) [E in Context Link 1, 2]

[F] Pediatric patients who have chronic liver disease with a Pediatric Model for End-Stage Liver Disease (PELD) score greater than 25 and have a need for mechanical ventilation or continuous renal replacement, ongoing gastrointestinal bleeding requiring 30 mL/kg or more of red cell transfusions within 24 hours, or a Glasgow coma score less than 10 may be registered as a status 1B for transplant, indicating an urgent need for liver transplant.(1) [F in Context Link 1]

[G] Hepatopulmonary syndrome is characterized by hypoxemia due to intrapulmonary vascular dilation in the setting of portal hypertension or cirrhosis.(24)(25) [G in Context Link 1, 2]

[H] Portopulmonary hypertension with a mean pulmonary artery pressure of greater than 35 mm Hg is associated with a high mortality rate post liver transplant.(25) [H in Context Link 1]

[I] Total hepatectomy with a temporary portacaval shunt may be required in severe hepatic trauma or primary liver allograft nonfunction immediately after transplant.(3)(30) [I in Context Link 1]

[J] Some patients may require simultaneous or interval kidney transplant.(51) [J in Context Link 1]

[K] See Clinical Indications for Procedure in this guideline. [K in Context Link 1]

[L] Many patients can be safely extubated immediately postoperatively or within a few hours of surgery.(53)(54) [L in Context Link 1]

[M] Effective antiemetic agents include 5-hydroxytryptamine receptor antagonists, dexamethasone, anticholinergic agents, dopamine antagonists, or neurokinin-1 receptor antagonists. The use of 2 agents for antiemetic prophylaxis is recommended.(55) [M in Context Link 1]

[N] Use Multimodal analgesia or individual analgesic agent as indicated. [N in Context Link 1]

[O] For very young patients (eg, infants), activity goals should be modified based on age and development. [O in Context Link 1, 2]

[P] Patients are discharged on immunosuppression.(6) [P in Context Link 1]

[Q] Patient is ambulatory or near baseline activity for age and development. [Q in Context Link 1]

[R] Some patients may have their hydration needs met via alternative means (eg, percutaneous endoscopic gastrostomy tube). [R in Context Link 1]

[S] Discharge instructions should be given in the patient's and caregiver's native language using trained language interpreters whenever possible.(67) [S in Context Link 1]

Definitions

Hemodynamic stability

- Hemodynamic stability, as indicated by **1 or more** of the following:
 - Hemodynamic abnormalities at baseline or acceptable for next level of care
 - Patient hemodynamically stable, as indicated by **ALL** of the following(1)(2)(3)(4)(5):
 - Tachycardia absent
 - Hypotension absent
 - No evidence of inadequate perfusion (eg, no myocardial ischemia)
 - No other hemodynamic abnormalities (eg, no Orthostatic hypotension)

References

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Hypotension

- Hypotension, as indicated by **ALL** of the following(1)(2)(3)(4):
 - Not patient baseline (eg, healthy adult with low SBP) or intentional therapeutic goal (eg, low SBP as treatment goal in heart failure)
 - Low blood pressure, as indicated by **1 or more** of the following:
 - SBP less than 90 mm Hg in adult or child 10 years or older[A]
 - Mean arterial pressure[A][B] less than 70 mm Hg in adult or child 10 years or older
 - Decrease in baseline mean arterial pressure of 25% or more,[A][B] with significant signs or symptoms due to lower blood pressure (eg, near syncope, syncope, chest pain)
 - SBP less than sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age[A]
 - SBP less than 70 mm Hg in infant 1 to 11 months of age[A]

References

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = 1/3 SBP + 2/3 DBP. A calculator is available at <https://www.mdcalc.com/mean-arterial-pressure-map>.

Hypotension absent

- Hypotension absent[A], as indicated by **1 or more** of the following(1)(2)(3)(4):
 - SBP greater than or equal to 90 mm Hg in adult or child 10 years or older
 - Mean arterial pressure[B] greater than or equal to 70 mm Hg in adult or child 10 years or older
 - Mean arterial pressure[B] at patient's baseline (eg, healthy adult with low SBP), at intentional therapeutic goal (eg, patient with heart failure), or acceptable for next level of care (eg, blood pressure stable and no significant signs or symptoms due to low blood pressure)
 - SBP greater than or equal to sum of 70 mm Hg plus twice patient's age in years in child 1 to 9 years of age
 - SBP greater than or equal to 70 mm Hg in infant 1 to 11 months of age

References

1. Jones D, Di Francesco L. Hypotension. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:657-64.
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4. Singh S, Holmes JF. Pediatric trauma. In: Walls RM, editor. Rosen's Emergency Medicine: Concepts and Clinical Practice. 10th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:2052-2066.e3.

Footnotes

- A. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

- B. The mean arterial pressure takes into account both systolic and diastolic blood pressure readings and is calculated as Mean Arterial Pressure (MAP) = $1/3 \text{ SBP} + 2/3 \text{ DBP}$.

Hypoxemia

- Hypoxemia, as indicated by **1 or more** of the following(1):
 - Patient without baseline need for supplemental oxygen with oxygen saturation (SaO₂) of less than 90% or arterial blood gas partial pressure of oxygen (PO₂) of less than 60 mm Hg (8.0 kPa) on room air[A][B]
 - Patient without baseline need for supplemental oxygen who now requires supplemental oxygen to keep SaO₂ greater than 89% or PO₂ greater than 59 mm Hg (7.9 kPa)[A]
 - Patient with baseline need for supplemental oxygen who now requires increased supplemental oxygen to maintain oxygenation at baseline or acceptable level

References

- Dezube R, Lechtzin N. Hypoxia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:669-675.
- Jamali H, et al. Racial disparity in oxygen saturation measurements by pulse oximetry: evidence and implications. Annals of the American Thoracic Society 2022;19(12):1951-1964. DOI: 10.1513/AnnalsATS.202203-270CME.
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Footnotes

- A. Pulse oximetry (SpO₂) may underestimate arterial saturation (SaO₂), especially in people with dark skin pigmentation, thick skin, cold body temperature, or who are wearing colored nail polish, and thus may not accurately detect hypoxemia.(2) Where appropriate, pulse oximetry should be measured after warming fingers or removing nail polish. Pulse oximetry results should be assessed within the context of the patient's clinical presentation, and performance of an arterial blood gas considered for a more accurate assessment of oxygenation if indicated.(2) Specific target values may require adjustment when care is delivered at high altitude.(1)(3)
- B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Hypoxemia absent

- Hypoxemia absent, as indicated by **1 or more** of the following(1):
 - Arterial oxygen saturation (SaO₂) of 90% or greater or arterial partial pressure of oxygen (PO₂) of 60 mm Hg (8.0 kPa) or greater on room air[A][B]
 - Oxygen requirements are stable and arranged for at next level of care.

References

- Dezube R, Lechtzin N. Hypoxia. In: McKean SC, Ross JJ, Dressler DD, Scheurer DB, editors. Principles and Practice of Hospital Medicine. 2nd ed. New York, NY: McGraw-Hill Education; 2017:669-675.
- Rojas-Camayo J, et al. Reference values for oxygen saturation from sea level to the highest human habitation in the Andes in acclimatised persons. Thorax 2018;73(8):776-778. DOI: 10.1136/thoraxjnl-2017-210598.

Footnotes

- A. Specific target values (ie, 90% or 60 mm Hg (8.0 kPa)) may require adjustment when care is delivered at high altitude.(2)
- B. Criteria based upon clinician-acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Multimodal analgesia

- Multimodal analgesia involves the utilization of 2 or more analgesic agents with different mechanisms of action in order to provide additive or synergistic pain control, while minimizing side effects and reliance on opioids.(1)(2)(3)

References

1. D'Souza RS, Johnson RL. Regional and multimodal treatments of perioperative pain. In: Benzon H, editor. Practical Management of Pain. 6th ed. Philadelphia, PA 19103-2899: Elsevier; 2023:355-373.
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No contraindications to liver transplant present

- No contraindications to liver transplant present, as indicated by **ALL** of the following(1)(2)(3)(4)(5)(6)(7):
 - No sustained increases of intracranial pressure greater than 50 mm Hg
 - Cerebral perfusion pressure not less than 40 mm Hg for more than 2 hours
 - No irreversible brain damage
 - No advanced HIV disease[A](6)
 - No uncontrolled systemic infection (eg, cholangitis, bacterial peritonitis)
 - No active alcohol or illicit substance use[B]
 - No advanced cardiac or pulmonary disease (eg, untreated coronary artery stenosis of 70% or greater, severe valvular disease, severe chronic lung disease)
 - No unaddressed barriers to ability to comply with immunosuppression protocol and follow-up requirements (eg, adequate social support)
 - No history of persistent nonadherence
 - No anatomic abnormalities precluding liver transplant
 - No active extrahepatic malignancy[C]
 - No hemangiosarcoma
 - No Niemann-Pick disease type C

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Footnotes

- A. HIV-positive liver transplant candidates should have CD4 counts greater than 100/mm³ (0.1 x10⁹/L) and plasma HIV RNA levels that are expected to be undetectable at the time of liver transplant.(6)
- B. The duration of the period of sobriety required prior to liver transplant is controversial with no universally accepted time period. The most commonly employed duration is 6 months. Because studies have shown that mortality is high in patients with severe alcoholic hepatitis who do not undergo transplant, and that patients undergoing earlier liver transplant for severe alcoholic hepatitis have similar survival and alcohol relapse rates as those who have completed a 6-month waiting period, shorter durations of abstinence may be appropriate.(8)(9)(10)(11)(12) A specialty guideline recommends against basing the liver transplant candidacy on a fixed abstinence interval.(13)
- C. Liver transplant candidates with a history of extrahepatic malignancy should have received definitive treatment with an adequate tumor-free survival period. The determination of what constitutes an adequate tumor-free survival period varies depending on the type of malignancy and efficacy of the treatment received.(6)

Orthostatic hypotension

- Orthostatic hypotension,[A][B] as indicated by **1 or more** of the following(1)(2)(3):
 - Fall in SBP of 20 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position
 - Fall in DBP of 10 mm Hg or more 1 to 3 minutes after patient sits or stands from recumbent position

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Footnotes

- A. Concomitant measurements of the heart rate are important to measure to help diagnose subtypes of orthostatic hypotension (eg, the lack of a compensatory increase in heart rate is typical of autonomic failure and an exaggerated tachycardia may be reflective of volume depletion). However, the heart rate is not a component of the definition of orthostatic hypotension which relies upon blood pressure alone.(1)(2)(3)
- B. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.

Social Determinants of Health Assessment

- Risk of poor health outcomes may be increased by the presence of **1 or more** of the following social determinants of health(1)(2)(3)(4):
 - Housing insecurity, as indicated by **1 or more** of the following:
 - Individual or caregiver's current living situation is **1 or more** of the following(5):
 - Does not have own housing (eg, staying in a hotel, shelter, or with others)
 - Has own housing (eg, house, apartment), but at risk of losing it in the future (ie, behind on rent or mortgage)
 - Has own housing (eg, house, apartment), but has lived in 3 or more places in past year
 - Current housing has **1 or more** of the following:
 - Electrical appliances (eg, stove, refrigerator) not working or unavailable
 - Insufficient heating or cooling
 - Insufficient ventilation
 - Lead paint or pipes
 - Mold
 - Pests (eg, bugs) or rodents
 - Smoke detectors not working or unavailable
 - Food insecurity, as indicated by **1 or more** of the following(6):
 - In the past year, individual or caregiver ran out of food and did not have money to buy more food.
 - In the past year, individual or caregiver worried that they would run out of food before they received money to buy more food.
 - Insufficient transportation, as indicated by **1 or more** of the following(7):

- In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of transportation.
- In the past year, individual or caregiver missed nonmedical activities, work, or could not get things needed for daily living due to lack of transportation.
- Insufficient utilities, as indicated by **1 or more** of the following(8):
 - Utilities (eg, electricity, water, gas, or oil) are currently shut off or unavailable.
 - In the past year, electric, water, gas, or oil company threatened to shut off services.
- Personal safety risk, as indicated by **2 or more** of the following(6):
 - Individual is sometimes or frequently physically hurt by another person (including family member).
 - Individual is sometimes or frequently insulted or talked down to by another person (including family member).
 - Individual is sometimes or frequently threatened with physical harm by another person (including family member).
 - Individual is sometimes or frequently screamed or cursed at by another person (including family member).
- Insufficient dependent care, as indicated by **1 or more** of the following:
 - In the past year, individual or caregiver was unable to work due to lack of dependent care.
 - In the past year, individual or caregiver was unable to work more (additional) hours due to lack of dependent care.
 - In the past year, individual or caregiver missed medical appointments or could not get medications due to lack of dependent care.
 - In the past year, individual or caregiver missed nonmedical activities (eg, school, church, social activity) due to lack of dependent care.
- Depression risk, as indicated by **ALL** of the following:
 - In the past 2 weeks, individual had little interest or pleasure in normal activities on at least several days.
 - In the past 2 weeks, individual felt down, depressed, or hopeless on at least several days.

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Tachycardia absent

- Tachycardia^{[A][B]} absent, as indicated by **1 or more** of the following(1)(2):
 - Heart rate less than or equal to 100 beats per minute in adult or child 6 years or older
 - Heart rate less than or equal to 115 beats per minute in child 3 to 5 years of age
 - Heart rate less than or equal to 125 beats per minute in child 1 or 2 years of age
 - Heart rate less than or equal to 130 beats per minute in infant 6 to 11 months of age
 - Heart rate less than or equal to 150 beats per minute in infant 3 to 5 months of age
 - Heart rate less than or equal to 160 beats per minute in infant 1 or 2 months of age

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Footnotes

- A. Criteria based upon clinician acquired numeric values (eg, vital signs, oxygen saturation) should be used if they are accurate reflections of the patient's condition. Transitory findings (eg, abnormal only upon initial emergency department intake or only one time out of multiple readings) that rapidly improve with no or minimal treatment usually do not reflect disease severity or risk for deterioration. This does not imply that an initial or one-time reading cannot ever be applicable. The goal is to separate erroneous or incidental findings from those that truly represent the patient's clinical picture.
- B. Interpretation of heart rate requires clinical judgment and consideration of several patient-specific factors, such as the patient's baseline heart rate, medications, and clinical impact. For example, an elderly patient on a beta-blocker medication with a baseline resting heart rate of 60 beats per minute may be clinically tachycardic at a heart rate of 94 beats per minute. Likewise, a patient who is upset, in pain, or nervous in the emergency department with a heart rate of 106 beats per minute may meet the technical definition of tachycardia, but this tachycardia (absent associated findings such as chest pain or hypotension) may not be clinically important. The numeric values included in this definition are provided to allow for consistency in terms of a technical definition of the term tachycardia. Whether a heart rate above or below the technical threshold is clinically meaningful is a matter of persistence, context, and clinical judgment.

Codes

ICD-10 Diagnosis: C22.0, C22.2, C22.3, C22.4, C22.7, C22.8, C22.9, C78.7, C7B.02, E72.53, E74.00, E83.01, E83.110, E85.2, E88.01, K70.30, K70.31, K70.40, K70.41, K72.00, K72.01, K72.10, K72.11, K72.90, K72.91, K73.1, K73.2, K73.8, K74.3, K74.4, K74.5, K74.60, K74.69, K75.4, K75.81, K76.0, K76.2, K76.81, P78.81, Q44.2, Q44.3, Q44.5, Q44.6, T86.42 [Hide]

ICD-10 Procedure: 0FY00Z0, 0FY00Z1, 0FY00Z2

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